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The Development of Project Monitoring Information System (Case Study: PT Tetapundi Prima Kelola)

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Abstract- PT Tetapundi Prima Kelola (TPK) is a small information technology company located in Jakarta established in 2012 to bridge the technology gap between Indonesia and other Asian countries, such as Hong Kong, Singapore, and Malaysia. The customers of this company come from various locations in or outside Jakarta capital city. Owing to several obstacles to monitor any particular project progress, this company needs a system to connect Technicians, Sales Department Staffs, and Manager of Sales Department for transforming the project progress data into project progress information. As a result, project-monitoring systems can be employed in this company, where web-based system can help the Sales Department staff to create project progress, which can be easily previewed by the Manager of Sales Department. The creation of database in this system can help the user to store data and information for the order details and project progress. Waterfall Model is used in developing this information system of project monitoring. The design of the system used is a notation of Unified Modeling Language (UML). In the implementation phase, system coding is done using PHP and MySQL, and blackbox testing is conducted to test the system. The outcome of this research is a system design that could help the Manager of Sales Department at PT TPK in monitoring a particular project progress.

I. INTRODUCTION

PT Tetapundi Prima Kelola (TPK) is a small information technology company established in Jakarta in the beginning of 2012 to bridge the technology gap between Indonesia and other Asian countries, such as Hong Kong, Singapore, and Malaysia. The customers of this company come from various locations in or outside Jakarta capital city. This company consists of six departments: Sales, Operational, Marketing, Human Resource, Accounting, and Warehouse; however, this research is focused on the Sales Department. The main rules and responsibilities of the Sales Department in this company are: receive customer's order, create order details documentation, input customer's details, and create project progress. Based on PT TPK business records, the order quantity is increasing every year, starting from early 2012 to late 2013. In early 2012, there were few orders in this company; however, by the end of 2013, there were more than 20 orders in a year. This company needs a system to control, manage, and also monitor business operations to achieve its business goals. For this, PT TPK needs a system that can help the Sales Department to represent the project progress, and as a result, the Manager of Sales Department could easily monitor the particular project progress.

Based on the background already explained, the problem identified is that the current system used is not efficient, due to some reasons, such as difficulty in previewing project progress details, vulnerable system for data or destruction, hard to control, and costing more money and effort. The main question that appears when conducting this research is how to develop information system of project monitoring at PT TPK that can manage customer's order details and project progress details, and display the updating of project progress in the report form. This research is focused on designing information systems of project monitoring that are limited to the Sales Department at PT TPK using data collection method and system development lifecycle method where PHP programming language, MySQL database management to create database, FusionChart data visualization, and some software, such as Adobe Dreamweaver CS6 and Notepad++ are used to develop the system.

The objective of this research is to develop an appropriate project-monitoring system at PT TPK that can help the Manager of Sales Department in previewing a particular project progress in graphical display.

II. LITERATURE REVIEW

A. Systems Development

The term of a system can be described as a particular process to capture, design, test, and implement a program or software application, including some internal customized system, internal development, the creation of database systems, or the acquisition of third-party developed software^[1]. In developing systems, there are seven important steps that must be followed by an organization: identify the problem, analyze and understand the problem, identify solution requirements and expectations, identify alternative solutions and choose the best course of action, design the chosen solution, implement the chosen solution, and evaluate the result^[2].

B. Information Systems

Historically, data refers to the facts concerning objects and events that could be recorded and stored on computer media^[3]. For example, in customer details database, the data may include different kinds of facts, such as customer name, address, email, and phone number. These particular examples are structured data that has data type, for example numeric,

character, and dates. Information is described as facts or details about a particular object that has been processed in a specific way where the user's knowledge will be increased^[4].

Systems as a group of irrelevant components that combine the function together to achieve the particular goals contain several components where each component has an interaction with each desired goal^[5]. Information system made up from two different words that create the collaboration between information and system refer to the collection of hardware, software, data, people, and also procedure that work together to create good quality of information^[6].

C. Project Monitoring

Project, done by an organization, group, company, and institution, is a temporary endeavor undertaken to create a specific product, service, or result^[7]. Project must have a certain specific purpose with a temporary length of time, and developed using progressive elaboration. Monitoring as an act to watch and check the object over the period of time means "the systematic and continuous assessment of the progress of a piece of work over time"^[8]. Monitoring provides the necessary information about project management and stakeholders to evaluate the project progress, identify trends, patterns, or deviations, keep project schedule, and measure progress towards expectation goals^[9]. The aspects that must be of concern in monitoring are timing, scope, main participant, process, and written outputs. Timing refers to the duration of the project and scope supports daily activities that influence the output of the project. The main participant in the project could be project staffs and project users and process of monitoring can be the regular meetings and interviews.

The fundamental principle of a monitoring system is to allow users to capture data, process, and disseminate information in a systematic way, where monitoring system enables one to measure the trends of various indicators based on the collection of data in a field^[10]. There are two types of monitoring: input monitoring and performance monitoring. The example of input monitoring is the administration of the distribution of textbooks to schools or an account of the delivery of in-service workshops to teachers; however, performance monitoring is directed to the processes in which a project is involved, to assure that these processes are indeed contributing to the expected output^[11]. Performance monitoring is applied for the system that would be developed by a researcher and used in PT TPK.

III. METHOD

A. Data Collection Method

Researchers use a variety of data collection methods to gather data and information required in developing information systems of project monitoring at PT TPK. There are three different kinds of data collection methods used in this research: library research, field study, and literature survey.

In library research, the researchers read several books, websites, and journal articles as references. Some references

used are: System Analysis and Design (Fifth Edition 2013) by Dennis, A., Wixom, B. H. & Roth, R. M.; Project Management (Second Edition 2000) edited by Richard H. Thayer; Systems Analysis & Design for the Global Enterprise (Seventh Edition 2007) by Lonnie D. Bentley and Jeffrey L. Whitten; Software Engineering (A Practitioner's Approach Seventh Edition 2010) by Roger S. Pressman; Practical Object-Oriented Design with UML (Second Edition 2003) by Mark Priestley.

In a field study, observations and interviews are conducted. Observation was conducted from October 30, 2013 to November 30, 2013 at PT TPK located in Jalan Cipinang Cempedak IV No. 7, Polonia, East Jakarta, 13340. The aim of the observation was to gather project-monitoring process operated in this company where the information retrieved from the observation is used to support the development of information system of project monitoring to PT TPK. To complete the data required in developing the system, interview sessions were also conducted during the observation session with some selected employees of this company. The employees provided their experience in using the current system to support in monitoring the project progress. Some of the questions asked during the interview session were: How the business process in this company operates?, Where the data and information of business process terminate?, What are the difficulties in this company when monitoring the project progress?, and Which group of employees currently have the responsibility to monitor the project progress?

Literature survey is conducted to support this research based on previous related researches. The comparison among several related researches about project monitoring is implemented, where some of the previous researches are: a systematic approach for monitoring and evaluating the construction of project progress, developing an information system for monitoring student's activity online collaborative learning, and monitoring in educational development projects, the development of a monitoring system, web-based information system for pollution monitoring of lake, and information system for monitoring environmental impacts of genetically modified organisms^{[12][13][14][15][16]}.

B. System Development Method

Waterfall model is one of the options for software development lifecycle model to view the activities that occurred during software development, and help the user to understand the basic activity and characteristics of the software^[17]. Waterfall model is used in developing project-monitoring systems at PT TPK. Four main phases in waterfall model are planning, analysis, design, and implementation^[18].

In the planning phase, business value of the information system of project monitoring at PT TPK is defined and research project plan is created. In the analysis phase, there are two stages: strategy analysis and system analysis. In strategy analysis, the current system is analyzed. In addition, analyzing the required system and recommended system is part of the system analysis phase.

In the design phase, design process is distinguished into five different processes: process, database, input and output, interface, and layout. In process design, the Unified Model Language (UML) tool is used to design information system development of project-monitoring process by three different phases, including create use case diagram, create activity diagram, and create sequence diagram. In database design, database that will be used to store data and information is created by six stages, including create potential object, create class diagram, optimize database, CRUD Matrix, database schema, and database specification. In input and output design, input and output of the recommended system is well defined. The input for this system will be data of order details and project progress; the output will be project progress percentage in graphical display, such as bar chart, pie diagram, and line chart. In interface design, the system interface based on user types is produced. In layout design, the interface design of the recommended system is captured and displayed.

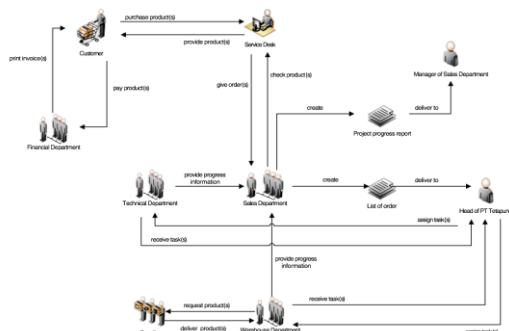
In the implementation phase, coding and testing are conducted. In coding, PHP programming language, MySQL as database management systems, FusionChart as data visualization application are the main tools used in developing the system. In the testing phase, the application that has been completely developed is being tested using blackbox testing type that focuses on the software functional requirements, and error in incorrect or missing function, application interface, data structure, or external database access, behavior or performance, and initiation and terminology.

IV. RESULTS AND DISCUSSION

A. Analysis

According to the researcher's analysis, the current system to monitor project progress is operating manually using *Microsoft Word* with the objective to focus on monitoring project progress, specifically for projects that operate outside Jakarta capital city. Business process in this company begins from the customers who order product. Once the Sales Department staff receive the order from the customer, they are responsible for creating the order details. The order detail will be used as data and information to monitor the specific project progress. Rich picture to present "mental assumptions about the relationship and interconnections" among the elements in the system is used to display the current system at PT TPK in Figure 4.1^[19].

Figure 4.1 Current systems at PT TPK rich picture



PIECES (Performance, Information, Economic, Control, Efficiency, Services) analysis is discovered in the current system. The feasibility of recommendation system is described in Table 4.1.

Table 4.1
PIECES ANALYSIS

Analysis Type	Current System Weaknesses	Recommendation System
Performance	Phone usage requires more time and big efforts; several mistakes may occur, for example mistyping from the Sales Department staff while inserting project progress data in <i>Microsoft Word</i> document as the user gets no warning for inappropriate data, and difficulty to preview the project progress details and percentage in graphical display.	Web-based project-monitoring system could reduce time and effort, because the user only needs to login into the system using his/her computer, diminish human error from staff who insert project data into recommended project-monitoring systems, because there will be warning to the user who inserts inappropriate data, and display details and percentage of project progress in graphical model (e.g. bar chart, pie diagram, and line graph).
Information	Mobile communication increases the time to retrieve project progress information, and the use of paper-based data storage will result in a vulnerable system for data or destruction.	Web-based technology can accelerate the delivery of information, and data can be stored and well-structured in the database provided.
Economic	Long-term use of phone between the Manager of Sales Department and Sales Department staff will be expensive.	Short-term use costs more money for developing web-based project-monitoring system; however, long-term use costs only system maintenance.
Control	Hard to control, because there are many people involved; as a result, some mistakes could occur.	Web-based system diminishes the number of people involved in project-monitoring system, because only the admin has the role and responsibility to input, update and delete project progress.
Efficiency	Communication between the Manager of Sales Department and Sales Department staff by phone is not cost-effective and is time-consuming.	Web-based project monitoring is more efficient, because it automatically connects to the related departments.
Service	Service process to support project-monitoring progress spends more time, because the process runs manually.	Shorter service because the entire information is deployed in a web-based system.

B. Design

In design process, the UML is used in developing the prototype for system design. The use case created for this system includes login, register new user, edit current user, delete current user, add customer, edit customer, delete customer, add product, edit product, delete product, input

order, edit order, preview order, delete order, update project progress, view project progress in bar chart, view project progress in pie diagram, view project progress in line graph, print project progress in bar chart, print project progress in pie diagram, print project progress in line graph, and logout.

Based on the researcher's analysis, the recommended system needs 11 tables in database, including user, employee, customer, order, product, project progress, permit letter, special, department, city, and province.

The data required as input has two forms, namely order details form and project progress form. The output of the system that will be displayed from data processing from database in this system is project progress in graphical display (including bar chart, pie diagram, and line graph).

The interface design for homepage system is displayed in Figure 4.2.

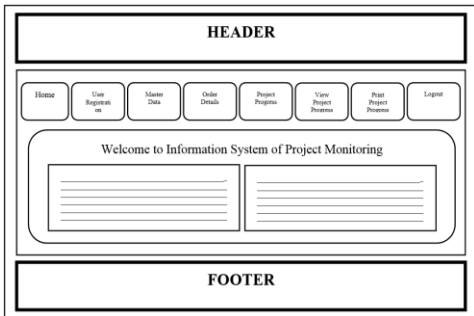


Figure 4.2 System interface design for the recommended system

The layout design for homepage system is displayed in Figure 4.3.



Figure 4.3 Layout design for the recommended system

C. Implementation

System blueprint previously designed is executed as a set of program codes using PHP programming language that will be implemented as project-monitoring system at PT TPK. Several tools that are used to develop project-monitoring system application are PHP, MySQL, XAMPP Apache HTTP Server, PHP MyAdmin, Adobe Dreamweaver, Internet Explorer, and Google Chrome.

In the testing stage, blackbox testing type is used to test the recommended project-monitoring system at PT TPK.

Table 4.2
LOGIN TESTING

Login data testing		
Actor	Admin and Manager	
Menu	Login	
Sub Menu	Login	
Action	Click login button	
Hint	Warning Error	Password or Username is incorrect
	Suitability Input	User is successfully login
Result	Warning	Appear
	Accept	Appear
Screen Capture		
Warning Error		
Password or Username is incorrect		

V. CONCLUSION

In developing information system of project monitoring, data collection and system development methods are used, where this recommended system might be implemented in PT TPK to support them in monitoring the progress of a particular project. The system is built using PHP programming language, MySQL database management, FusionChart data visualization application, and some additional supported software.

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